

SESSION-7:

Tasks

1.

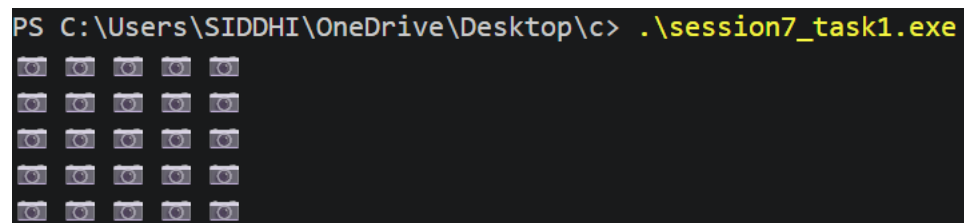
Use nested for loops to print a grid of emojis representing a 5x5 Instagram post feed, where each cell shows a 📷 symbol.

```
#include <stdio.h>
```

```
int main() {  
    for (int i = 0; i < 5; i++) {  
        for (int j = 0; j < 5; j++) {  
            printf("📷 ");  
        }  
        printf("\n");  
    }  
}
```

```
return 0;
```

```
}
```



```
PS C:\Users\SIDDHI\OneDrive\Desktop\c> .\session7_task1.exe  
📷 📷 📷 📷 📷  
📷 📷 📷 📷 📷  
📷 📷 📷 📷 📷  
📷 📷 📷 📷 📷  
📷 📷 📷 📷 📷
```

2.

Build a right-angled triangle pattern using nested loops, where each row displays increasing numbers starting from 1, similar to how a leaderboard on a gaming app shows rank numbers.

```
#include <stdio.h>
```

```
int main() {  
    for (int i = 1; i <= 5; i++) {  
        for (int j = 1; j <= i; j++) {
```

```

        printf("%d ", j);
    }
    printf("\n");
}

return 0;
}
PS C:\Users\SIDDHI\OneDrive\Desktop\c> gcc session7_task2.c -o session7_task2
PS C:\Users\SIDDHI\OneDrive\Desktop\c> .\session7_task2.exe
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

3.

Create a pattern that prints a pyramid of stars (*) with 6 rows, centered like the loading animation you see on BookMyShow when a page is loading.

Hint: Use spaces to align the stars in the center for each row.

```
#include <stdio.h>
```

```

int main() {
    for (int i = 1; i <= 6; i++) {
        for (int j = 1; j <= 6 - i; j++) {
            printf(" ");
        }

        for (int j = 1; j <= 2 * i - 1; j++) {
            printf("*");
        }

        printf("\n");
    }

    return 0;
}

```

```
}
```

```
PS C:\Users\SIDDHI\OneDrive\Desktop\c> gcc session7_task3.c -o session7_task3
PS C:\Users\SIDDHI\OneDrive\Desktop\c> .\session7_task3.exe
*
***
*****
*****
*****
*****
*****
```

4.

Write code using nested loops to print a pattern of alternating 0s and 1s in a grid, like the checkered background seen in some Spotify playlist covers (e.g., for a 4x4 grid, alternate 0 and 1 in each cell).

```
#include <stdio.h>
```

```
int main() {
    for (int i = 0; i < 4; i++) {
        for (int j = 0; j < 4; j++) {
            printf("%d ", (i + j) % 2);
        }
        printf("\n");
    }

    return 0;
}
```

```
PS C:\Users\SIDDHI\OneDrive\Desktop\c> gcc session7_task4.c -o session7_task4
PS C:\Users\SIDDHI\OneDrive\Desktop\c> .\session7_task4.exe
0 1 0 1
1 0 1 0
0 1 0 1
1 0 1 0
```

5.

Modify your pyramid pattern code to accept the number of rows as user input, so the user can set the height of the pyramid before printing.

```
#include <stdio.h>
```

```
int main() {
```

```
    int rows;
```

```
    printf("Enter the number of rows: ");
```

```
    scanf("%d", &rows);
```

```
    for (int i = 1; i <= rows; i++) {
```

```
        for (int j = 1; j <= rows - i; j++) {
```

```
            printf(" ");
```

```
        }
```

```
        for (int j = 1; j <= 2 * i - 1; j++) {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

```
PS C:\Users\SIDDHI\OneDrive\Desktop\c> gcc session7_task5.c -o session7_task5.exe
```

```
PS C:\Users\SIDDHI\OneDrive\Desktop\c> .\session7_task5.exe
```

```
Enter the number of rows: 3
```

```
 *
***
*****
```